

Dashboard Filters

Select Year

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

2025

Select Month

December

November

October

September

August

July

June

May

April

Select Season

Winter

Autumn

Summer

Spring

Weekend / Weekday

Weekend

Weekday

Peak Period

Off Peak

Evening Peak

Morning Peak



Toronto Island Ferry Analytics Dashboard

Real-Time Ferry Ticket Sales & Redemption Analytics

This dashboard provides comprehensive operational analytics for the Toronto Island Ferry system using historical ticket sales and redemption data.



Executive KPI Dashboard

 Total Ticket Sales	 Total Redemptions	 Net Passenger Movement	 Utilization Rate
12,972,051	12,785,293	186,758	98.56%

 Peak Hour : 12:00	 Peak Day : Saturday	 Peak Month : August
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Dataset Overview

Rows	Columns	Years Covered
261,538	13	11

> Preview Dataset

Project Information

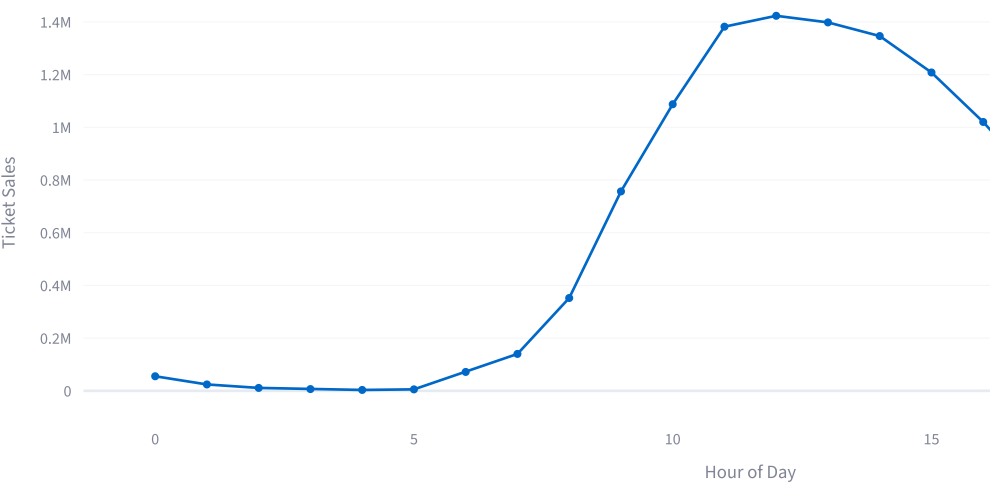
Project Name	Real-Time Ferry Ticket Sales and Redemption Analytics for Toronto Island Park
Domain	Data Analytics
Tools Used	Python Streamlit Pandas NumPy Matplotlib Seaborn Plotly SciPy
Dataset	Toronto Island Ferry Ticket Sales & Redemption

 Part 1 Loaded Successfully



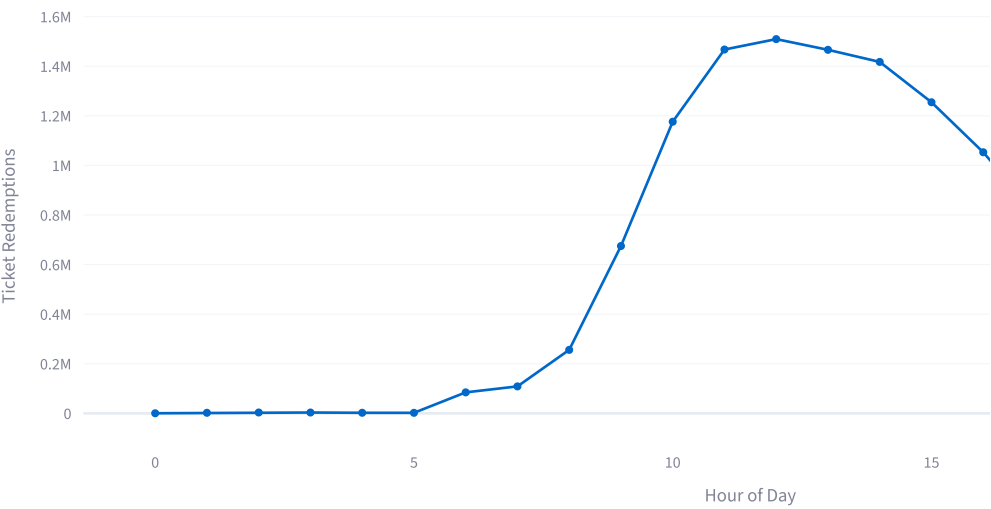
Hourly Analytics

Hourly Ticket Sales Trend



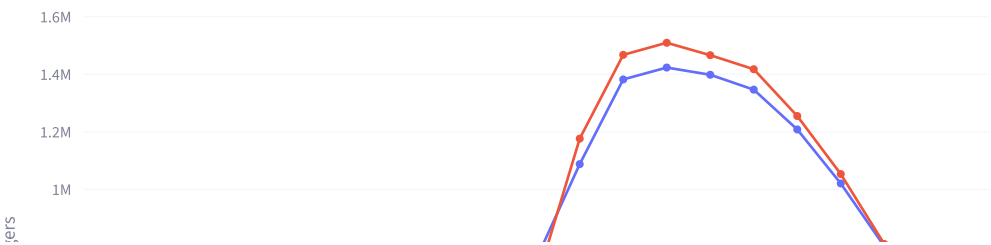
✦ **Insight:** Passenger ticket sales increase during the morning, reach their highest levels around midday, and gradually decline toward the evening, highlighting the busiest operating hours.

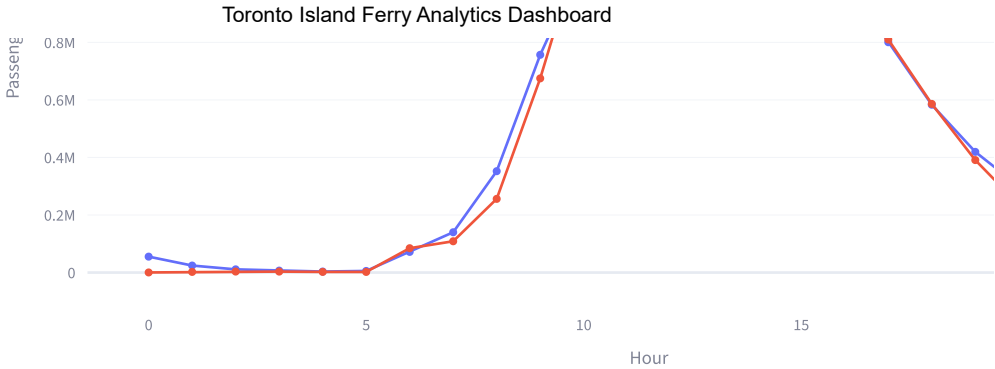
Hourly Ticket Redemption Trend



✦ **Insight:** Ticket redemption closely follows ticket sales, indicating consistent passenger utilization throughout the day.

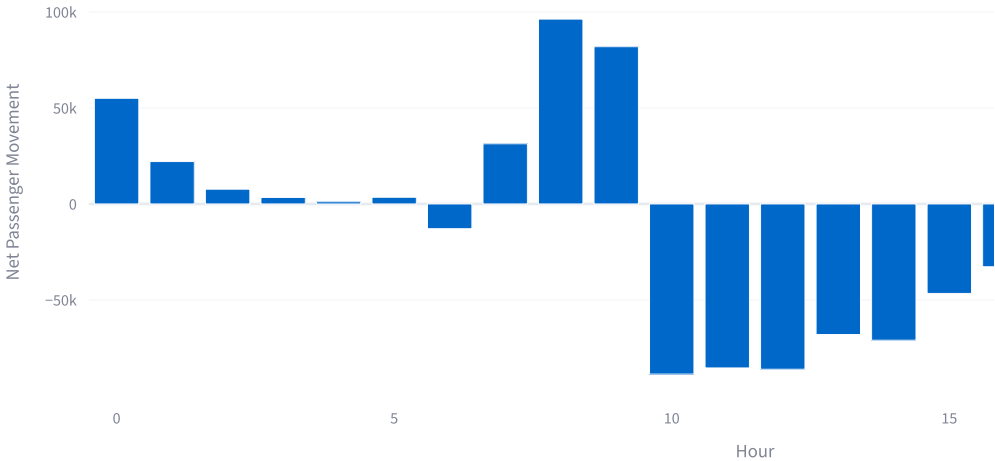
Hourly Sales vs Redemption Comparison





Insight: Sales and redemption patterns remain closely aligned throughout the day, demonstrating stable operational performance.

Hourly Net Passenger Movement



Insight: Net passenger movement varies across the day, providing valuable information for optimizing ferry capacity and terminal operations.



Top 10 Peak Demand Hours

Hour	Sales Count
12	
13	
11	
14	
15	
10	
16	
17	
9	
18	



Hourly Analysis Summary

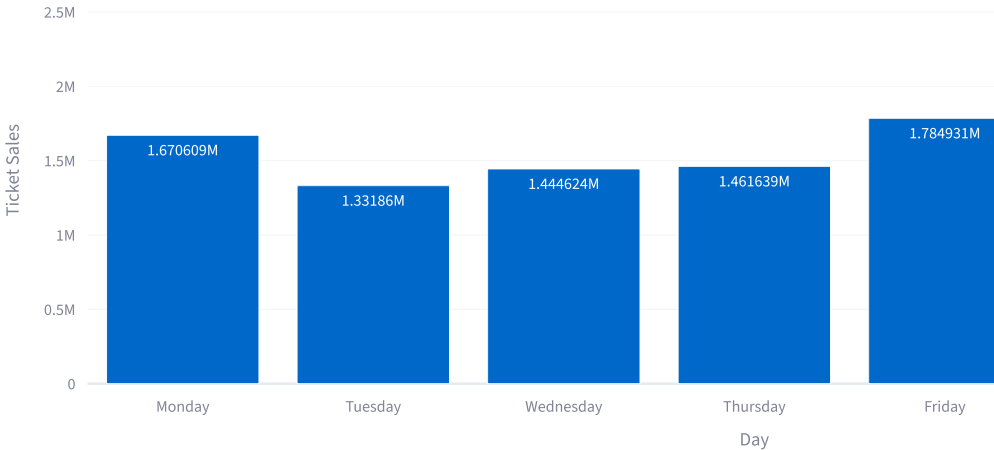


Hourly Analysis Loaded Successfully



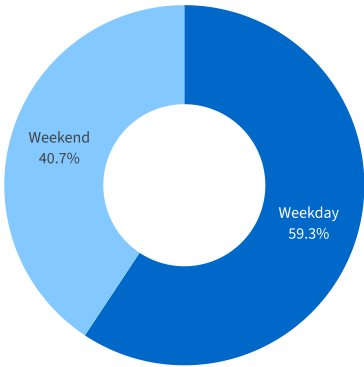
Daily & Weekly Analysis

Daily Ticket Sales Analysis



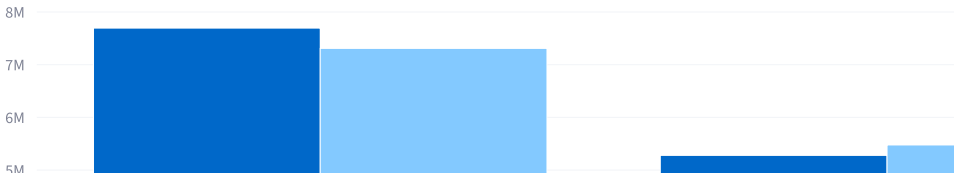
✦ **Insight:** Saturday records the highest passenger demand, indicating greater ferry utilization compared to other days of the week.

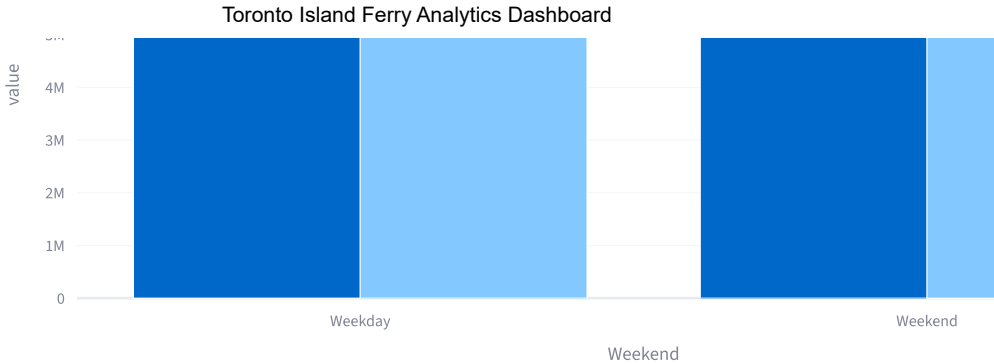
Weekend vs Weekday Contribution



✦ **Insight:** Weekend demand contributes significantly to total passenger activity, highlighting the importance of optimized weekend ferry operations.

Weekend vs Weekday Sales & Redemption





Insight: Ticket sales and redemption remain closely aligned across weekdays and weekends, demonstrating consistent passenger utilization.

Daily Passenger Summary

Day	Sales Count	Redemption
Monday	1670609	
Tuesday	1331860	
Wednesday	1444624	
Thursday	1461639	
Friday	1784931	
Saturday	2727394	
Sunday	2550994	

Highest Demand Day

Saturday

Total Daily Sales

12,972,051

Average Daily Sales

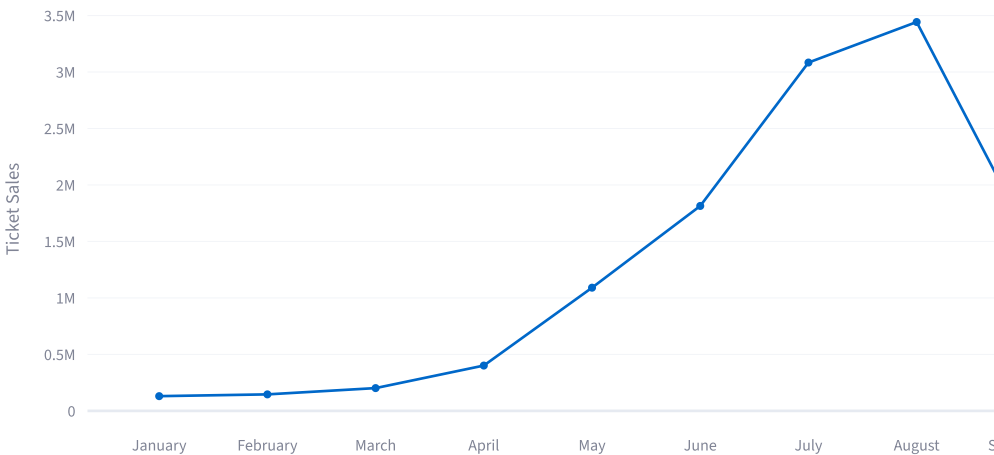
1,853,150

> Daily Analysis Summary

Daily & Weekend Analysis Completed

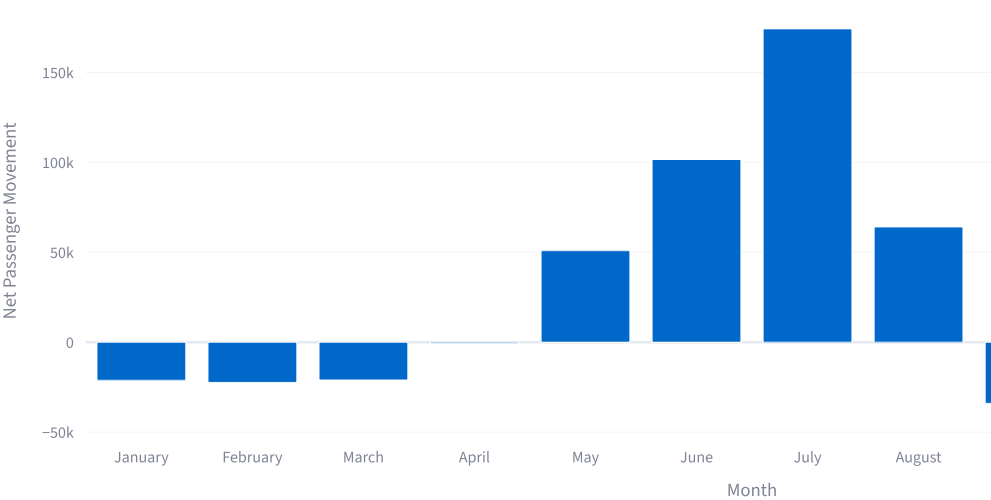
Monthly & Quarterly Analytics

Monthly Ticket Sales Trend



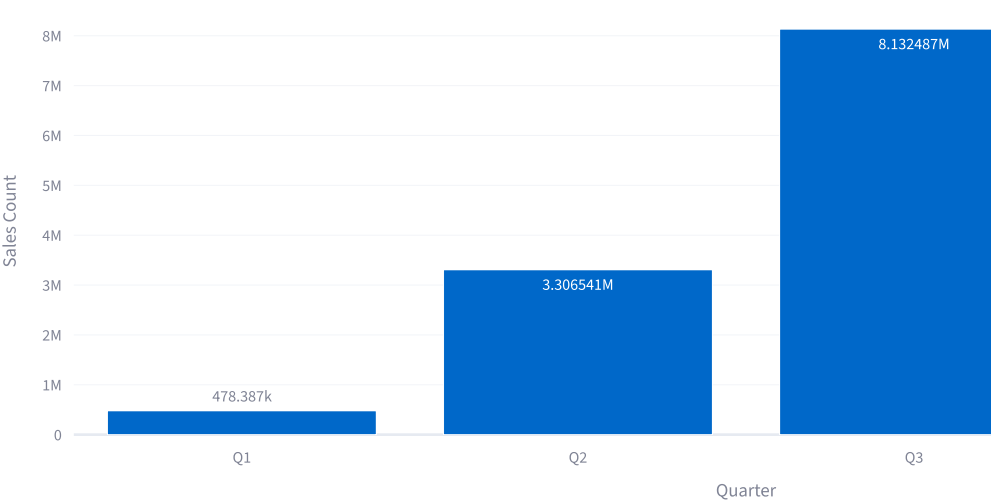
✦ **Insight:** August records the highest passenger demand, while January records the lowest demand, highlighting seasonal travel behaviour.

Monthly Net Passenger Movement



✦ **Insight:** Monthly passenger movement helps identify periods of high inflow and supports effective resource planning.

Quarter-wise Ticket Sales



✦ **Insight:** Q3 contributes the highest passenger demand, indicating increased travel activity during this quarter.

📄 Monthly Summary

Month	Sales Count	Redemption
January	130455	
February	146179	
March	201753	
April	401395	
May	1090751	
June	1814395	
July	3083858	
August	3442762	
September	1605867	
October	638215	

🏆 Best Month

August

📉 Lowest Demand Month

January

📊 Best Quarter

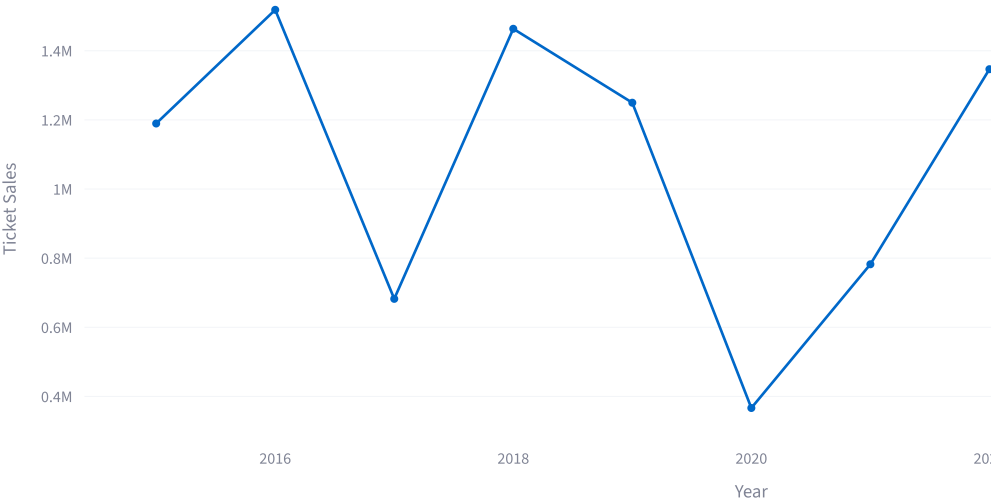
Q3

> 💡 Monthly Business Insights

✅ Monthly & Quarterly Analysis Completed

Yearly Analytics

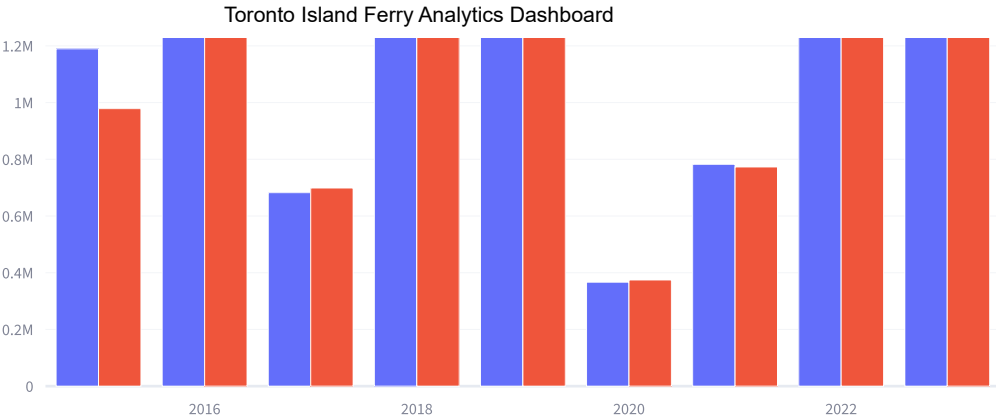
Yearly Ticket Sales Trend



🏆 Highest passenger demand was recorded in 2016.

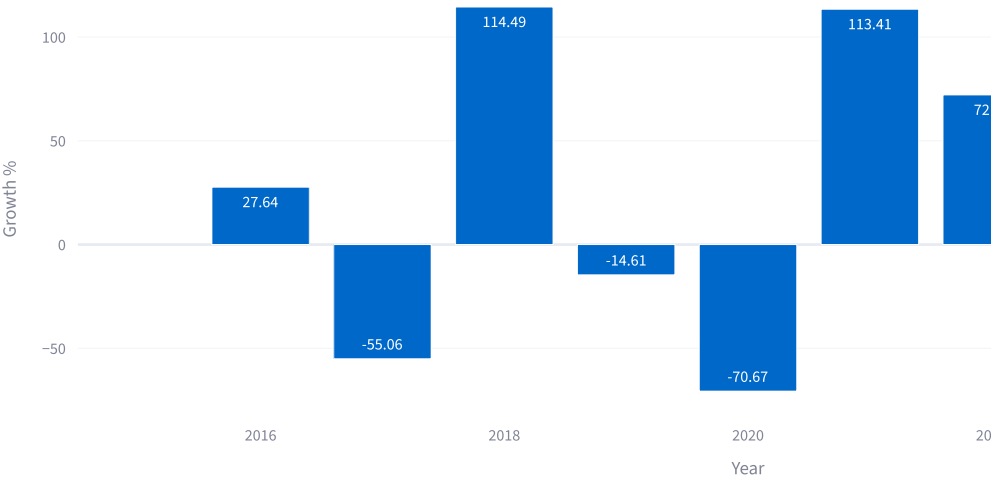
Year-wise Sales vs Redemption





📌 Ticket sales and redemption remain highly consistent across all years, demonstrating efficient ferry utilization.

Year-over-Year Sales Growth



📌 Positive growth indicates increasing passenger demand, while negative growth reflects temporary reductions in ferry utilization.

Years Covered	Best Year	Highest Sales	Average Yearly Sales
11	2016	1,518,428	1,179,277

📄 Yearly Summary

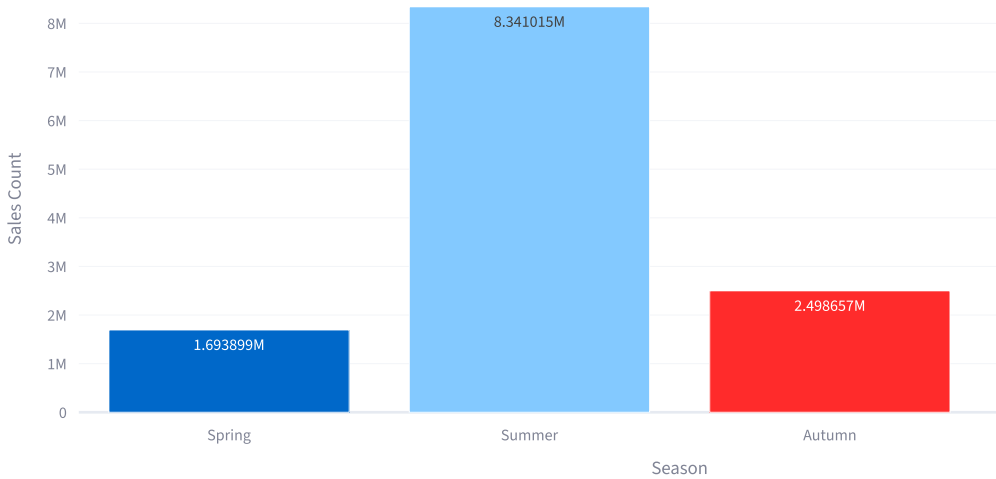
Year	Sales Count	Redemption Count	Net
2016	1518428	1425779	
2017	682346	698861	
2018	1463589	1493560	
2019	1249725	1278505	
2020	366606	374546	
2021	782368	773040	
2022	1346659	1349778	
2023	1491473	1502883	
2024	1373563	1390358	
2025	1507674	1519276	

> 💡 Yearly Business Insights

✅ Yearly Analytics Completed

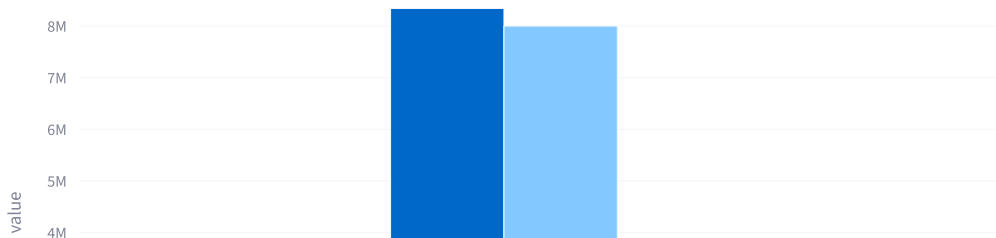
☁️ Seasonal & Peak Period Analytics

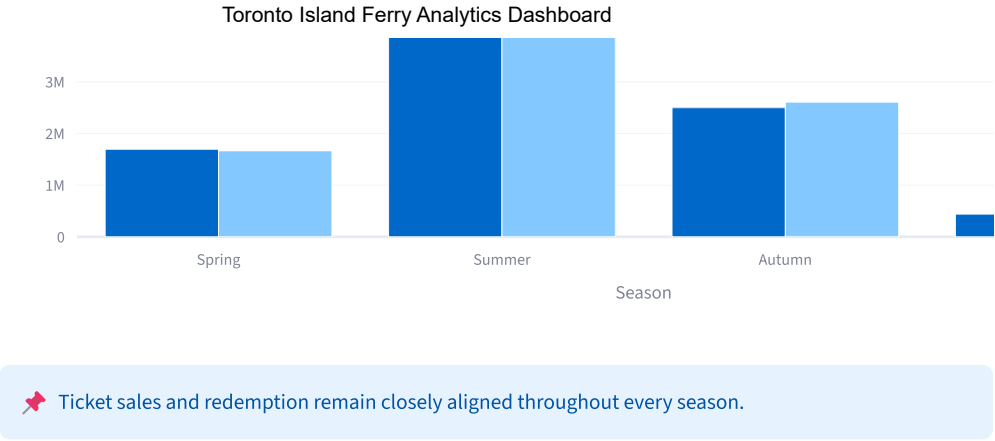
Seasonal Ticket Sales Analysis



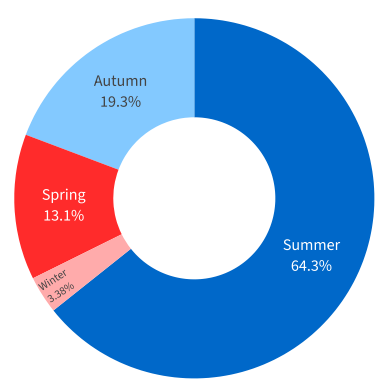
🏆 Highest passenger demand occurs during Summer.

Seasonal Sales vs Redemption

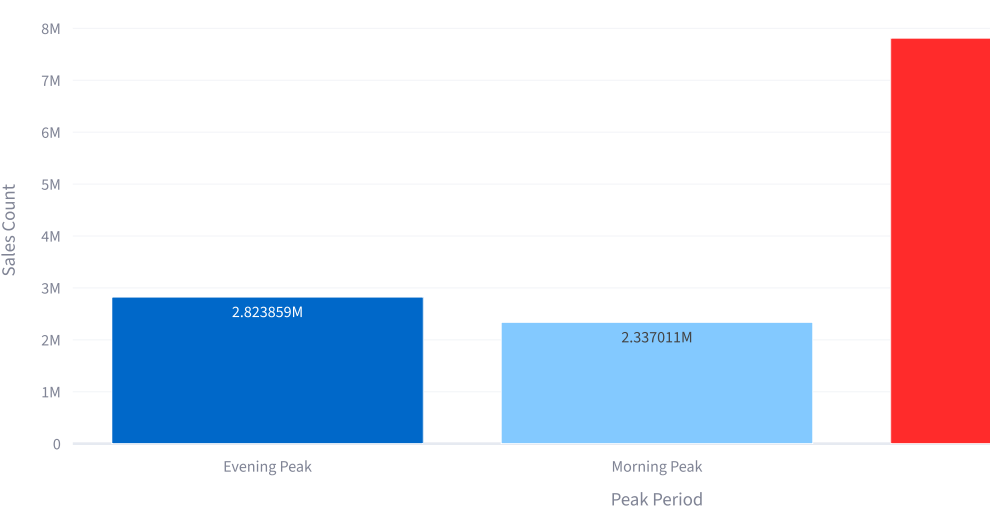




Seasonal Contribution to Total Sales

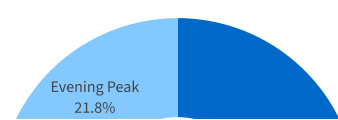


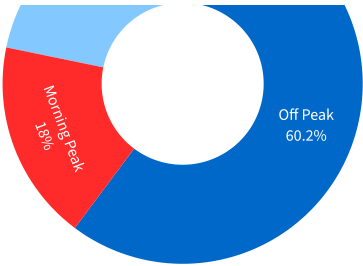
Peak vs Off-Peak Passenger Demand



✦ Passenger demand varies considerably across different operating periods, highlighting opportunities for optimized scheduling.

Peak Period Contribution





Best Season

Lowest Season

Peak Categories

Season Categories

Summer

Winter

3

4

 Seasonal Summary

Season	Sales Count	Redemption Count	N
Spring	1693899	1664773	
Summer	8341015	8000961	
Autumn	2498657	2606380	
Winter	438480	513179	

>  Seasonal Business Insights

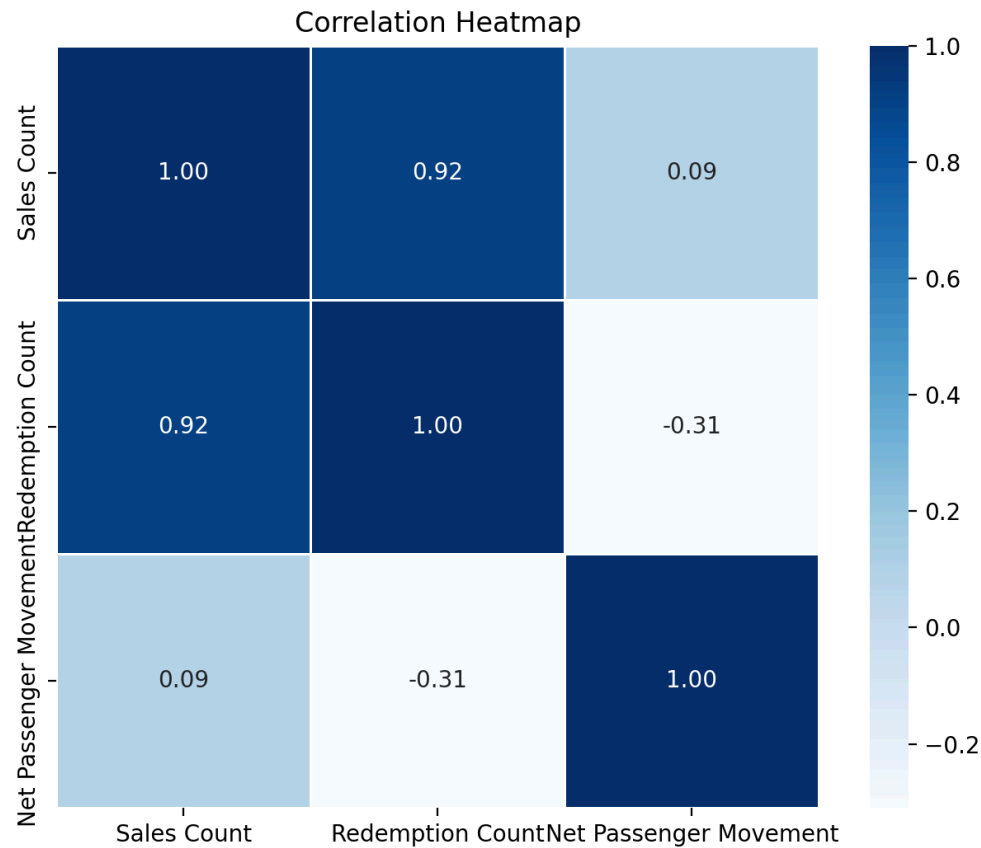
 Seasonal Analytics Completed

 Statistical Analysis

Correlation Matrix

	Sales Count	Redemption Count	Net F
Sales Count	1.000000	0.918163	
Redemption Count	0.918163	1.000000	
Net Passenger Movement	0.092997	-0.309100	

Correlation Heatmap



✦ Strong positive correlation between Ticket Sales and Ticket Redemption indicates highly consistent passenger utilization.

Pearson Correlation Test

Metric	Value
Correlation Coefficient	0.9182
P-value	0.0000

✔ Statistically Significant Relationship Found

Correlation Coefficient : 0.9182

P-value : 0.0000000000

Interpretation:

The Pearson correlation coefficient indicates a very strong positive relationship between ticket sales and ticket redemption.

Correlation	P-value	Relationship
0.9182	0.000000	Strong Positive

> 📊 Statistical Insights

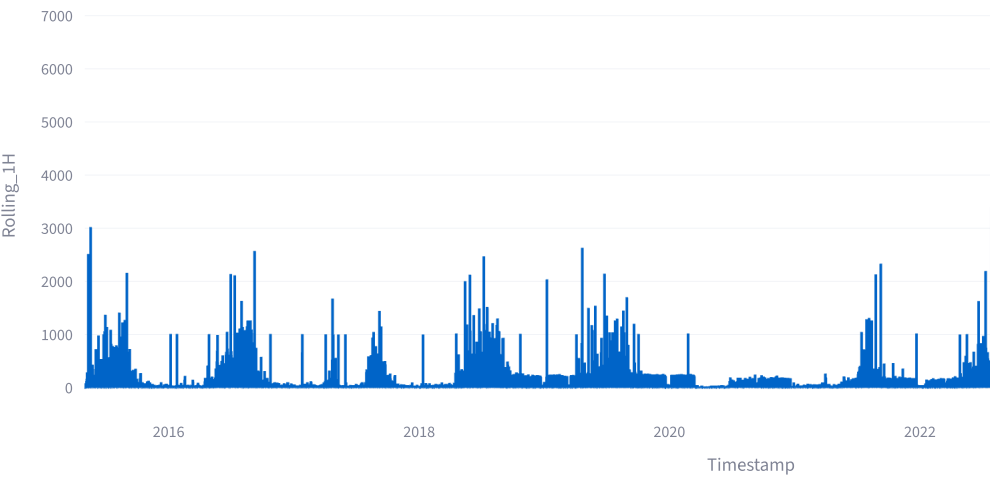
✔ Correlation Analysis Completed



Trend & Distribution Analysis

1-Hour Rolling Average

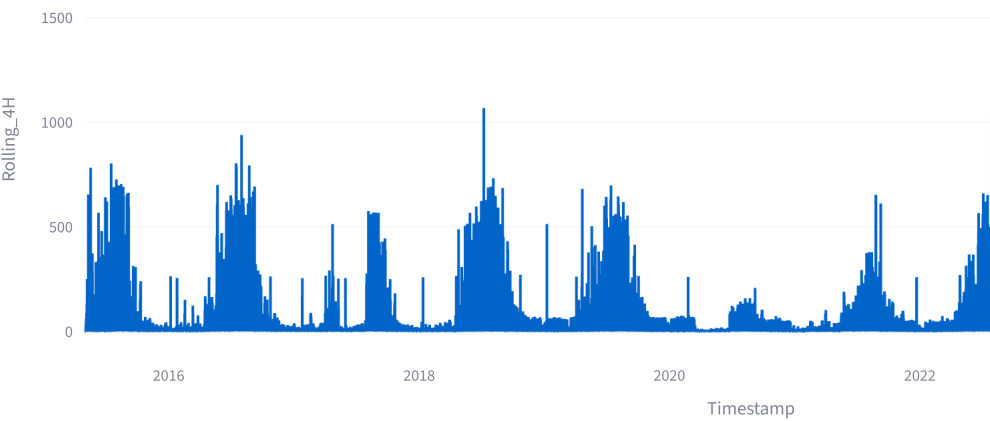
1-Hour Rolling Average of Ticket Sales



 **Insight:** The 1-hour rolling average smooths short-term fluctuations while preserving immediate passenger demand trends.

4-Hour Rolling Average

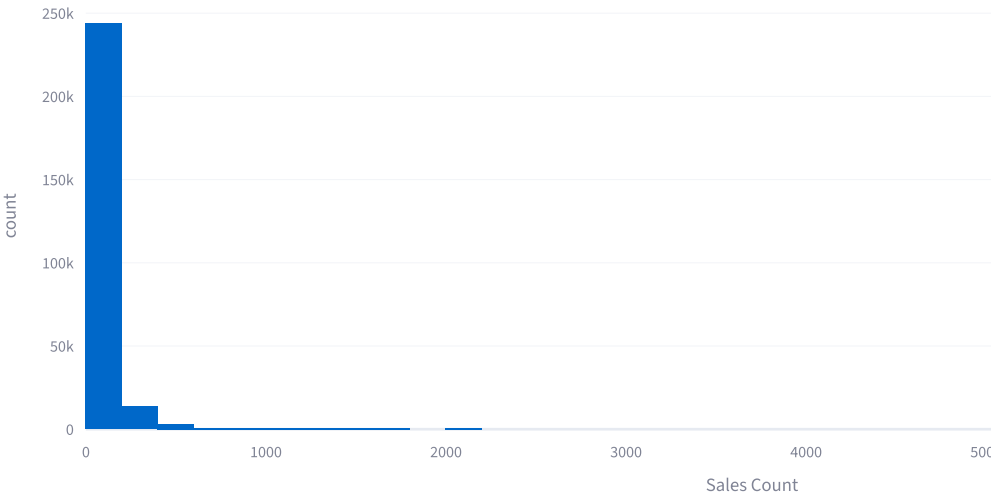
4-Hour Rolling Average of Ticket Sales



 **Insight:** The 4-hour rolling average highlights long-term demand behavior by reducing short-term variability.

Ticket Sales Distribution

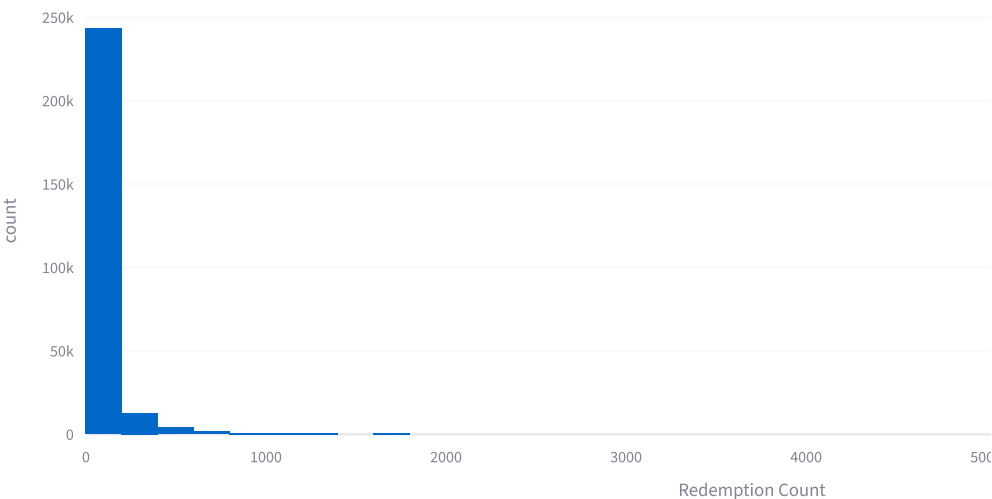
Distribution of Ticket Sales



✦ **Insight:** Most observations fall within lower ticket sales ranges, while a few high-demand periods create a positively skewed distribution.

Ticket Redemption Distribution

Distribution of Ticket Redemptions



✦ **Insight:** Redemption counts follow a distribution similar to ticket sales, confirming consistent passenger utilization.

Demand Volatility

Metric	Value
Average Sales	
Median Sales	
Standard Deviation	
Maximum Sales	
Minimum Sales	

✦ **Insight:** Standard deviation measures passenger demand variability. Higher values indicate larger fluctuations across different operating periods.

Average Sales	Median Sales	Std. Deviation	Maximum Sales
49.60	13.00	99.86	7,229

>  Statistical Summary

 Trend & Distribution Analysis Completed



Advanced Business Intelligence



Sales Count Outlier Detection

Sales Count Outlier Detection



 High-value observations represent genuine peak passenger demand rather than data quality issues.



Redemption Count Outlier Detection

Redemption Count Outlier Detection



✦ Redemption outliers correspond to exceptionally high passenger activity and assist in operational planning.

🏆 Top 10 Ticket Sales

Timestamp	Sales Count
2023-08-15 20:15:00	
2023-10-10 14:00:00	
2024-06-19 19:00:00	
2025-09-01 17:30:00	
2024-08-05 22:00:00	
2024-09-07 17:00:00	
2025-09-24 22:30:00	
2025-07-17 23:15:00	
2023-07-05 21:15:00	
2022-08-15 10:30:00	

🏆 Top 10 Ticket Redemptions

Timestamp	Redemption Count
2023-08-15 20:15:00	
2023-10-10 14:00:00	
2024-06-19 19:00:00	
2025-09-01 17:30:00	
2024-08-05 22:00:00	
2024-09-07 17:00:00	
2025-09-24 22:30:00	
2025-07-17 23:15:00	
2023-07-05 21:15:00	
2022-08-15 10:30:00	

📅 Top 10 Peak Demand Dates

Timestamp	Sales Count	Redempti
2023-08-15 20:15:00	7229	
2023-10-10 14:00:00	5518	
2024-06-19 19:00:00	5342	
2025-09-01 17:30:00	5218	
2024-08-05 22:00:00	4995	
2024-09-07 17:00:00	4974	
2025-09-24 22:30:00	4015	
2025-07-17 23:15:00	3701	
2023-07-05 21:15:00	3634	
2022-08-15 10:30:00	3408	

Executive Business Intelligence

Highest Demand

-  Year : 2016
-  Season : Summer
-  Month : August
-  Hour : 12:00
-  Day : Saturday

Statistical Highlights

- ✓ Correlation : 0.9182
- ✓ P-value : 0.00000000
- ✓ Utilization : 98.56%
- ✓ Peak Quarter : Q3

Business Recommendations

- ✓ Increase ferry frequency during peak operating hours.
- ✓ Deploy additional staff during weekends and summer.
- ✓ Utilize predictive analytics for demand forecasting.
- ✓ Monitor KPIs through real-time dashboards.
- ✓ Optimize resource allocation using historical demand.
- ✓ Integrate weather and holiday data into planning.
- ✓ Improve passenger experience through reduced waiting time.
- ✓ Adopt data-driven decision making for long-term planning.

Download Center

↓ Download Filtered Dataset

↓ Download Summary Statistics

Toronto Island Ferry Analytics Dashboard

Developed by: Satwik Srivastava

Internship Domain: Data Analytics

Tools Used

- Python
- Streamlit
- Pandas
- NumPy
- Plotly
- Matplotlib
- Seaborn

- SciPy

Dashboard Features

- ✓ Executive KPIs
- ✓ Interactive Filters
- ✓ Exploratory Data Analysis
- ✓ Statistical Analysis
- ✓ Business Intelligence
- ✓ Download Center
- ✓ Business Recommendations

 Dashboard Loaded Successfully